Getting Started

Once you have cloned this repository follow the coming instructions to perform a quick start to your first simulation with ARC.

Step 1) Navigate to /Evolution_Schemes/examples/sod_shock_tube_cpu.jl

Step 2) Once inside the Julia file go down to the Parameters and Grid for cpu section, you should see the following

```
# === Parameters & Grid for cpu ===
const γ = 4/3
N       = 4096
grid    = collect(range(-10.0, 10.0; length=N))
Δx      = (grid[end] - grid[1]) / N
CFL     = 0.3
t_final = 10.0

# === Initial Conditions ===
ρL, uL, PL = 1.0, 0.0, 1.0
ρR, uR, PR = 0.125, 0.0, 0.1



run_sod_shock_tube_cpu(ρL, uL, PL, ρR, uR, PR, grid, Δx, CFL, t_final)
```

Step 3) Set these parameters to your liking and run the simulation in the Julia terminal or within your IDE.

Description of each parameter

$\gamma \rightarrow$ Adiabatic constant

$N \rightarrow$ number of grid points

$grid \rightarrow$ the points used in the simulation. In the image above, -10 and 10 are the physical boundaries of the simulation

$\Delta x \rightarrow$ spacing between grid points

$CFL \rightarrow$ The Courant time step condition

$t_{final} \rightarrow$ The final time that you want to run the simulation till (to achieve full time, you may need to adjust the frames from 100 to more if you want to simulate the full t_{final} duration (the image below shows you where to find the spot to change it

```
# === Time Loop with Animation ===
@gif for frame = 1:100
    if t >= t_final
        break
    end

    #U = lax_friedrichs_step!(U, dt, Δx)
    #U = richtmyer_step!(U, dt, Δx)
    W,U = riemann_step!(W, grid, t)
    t += dt
    dt = compute_dt(U, Δx, CFL)
    step += 1
```

$\rho_L, u_L, p_L \rightarrow$ Left state of your system

$\rho_R, u_R, p_R \rightarrow$ Right state of your system